

Homework 1

Arjun Mehta Linear Algebra Sep 17, 2026 GRADE: 31.5/40 (Prof. R. Castellano)

Problem 1

(a) Final answer: 24

[PROF: -0.5 E1 Bare answer; $2u - v$ and the expanded dot product are not shown.]

(b) $(1/2) w = [-2, 1, 3]^T$

$$v + (1/2) w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T$$

$$\|v + (1/2) w\| = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3} \approx 5.20$$

Final answer: $3\sqrt{3}$

[PROF: ✓]

(c) $\cos \theta = (u \cdot w) / (\|u\| \|w\|)$

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$\|u\| = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$\|w\| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$\cos \theta = 8 / (\sqrt{14} \cdot 2\sqrt{14}) = 8 / 28 = 2/7$$

$$\theta = \arccos(2/7) \approx 73^\circ$$

Final answer: $\theta \approx 73^\circ$

[PROF: -1 E8 Angle rounded to 73° ; the problem requires two decimals, $\theta \approx 73.40^\circ$.]

[PROF: Problem 1 3.5/5]

Problem 2

(a) AB : undefined, because A has 3 columns and B has only 2 rows ($3 \neq 2$).

BA is 2×3 since B is 2×3 and A is 3×3 .

row 1: $[1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(4), 1(0) + 0(-2) + (-2)(1)] = [2, -9, -2]$

row 2: $[3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 4, -1]$

$BA = [2 \ -9 \ -2]$

$[7 \ 4 \ -1]$

[PROF: ✓]

(b) $Av = [2(1) + (-1)(-2) + 0(3), 1(1) + 3(-2) + (-2)(3), 0(1) + 4(-2) + 1(3)]^T = [4, -11, -5]^T$

vA : undefined, v is 3×1 and A is 3×3 , so 1 column cannot meet 3 rows.

Final answer: $Av = [4, -11, -5]^T$, vA undefined

[PROF: ✓]

(c) Aw : undefined, A is 3×3 and w is 1×3 , so 3 columns cannot meet 1 row.

$wA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)] = [4, -6, -1]$

[PROF: -0.5 N6 Final answer is neither boxed nor labeled Final answer.]

(d) vw is 3×1 times 1×3 , so it is 3×3 :

$vw = [2 \ 0 \ -1]$

$[-4 \ 0 \ 2]$

$[6 \ 0 \ -3]$

wv is 1×3 times 3×1 , so it is 1×1 : $2(1) + 0(-2) + (-1)(3) = -1$. $wv = [-1]$

[PROF: ✓]

(e) $A^2 = A \text{ times } A$:

row 1: $[2(2) + (-1)(1) + 0(0), 2(-1) + (-1)(3) + 0(4), 2(0) + (-1)(-2) + 0(1)] = [3, -5, 2]$

row 2: $[1(2) + 3(1) + (-2)(0), 1(-1) + 3(3) + (-2)(4), 1(0) + 3(-2) + (-2)(1)] = [5, 0, -8]$

row 3: $[0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 16, -7]$

$A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$

[PROF: ✓]

(f) B^2 undefined

[PROF: -0.5 E5 Undefined is stated with no dimension reason; B is 2×3 , so BB would need $3 = 2$.]

[PROF: Problem 2 5/6]

Problem 3

$$(a) \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_2 \rightarrow -R_2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 10 & | & 10 \end{bmatrix}$$

$$R_3 \rightarrow (1/10)R_3: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & | & 1 \end{bmatrix}$$

$x_3 = 1, x_2 = -4 + 3(1) = -1, x_1 = -1 - 2(-1) + 1 = 2$. One solution only.

Final answer: $x = [2, -1, 1]^T$

[PROF: ✓]

$$(b) \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 & 1 & | & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2: \begin{bmatrix} 1 & 0 & -7 & | & 8 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

There is no pivot in column 3, so z is free.

Final answer: z is free, so there are infinitely many solutions

[PROF: -1 E4 Free variable named but no parametric form; write $[x, y, z]^T = [8, -2, 0]^T + t[7, -3, 1]^T, t \in \mathbb{R}$.]

[PROF: Problem 3 5/6]

Problem 4

(a) Let x_1 = mass of Alloy 1 used, in kg; x_2 = mass of Alloy 2, in kg; x_3 = mass of

Alloy 3, in kg. The 100 kg of product holds 36 kg copper, 44 kg zinc, 20 kg nickel,

and none of the metal is lost, so in = out for each metal:

copper: $0.5x_1 + 0.2x_2 + 0.4x_3 = 36$

zinc: $0.3x_1 + 0.7x_2 + 0.2x_3 = 44$

nickel: $0.2x_1 + 0.1x_2 + 0.4x_3 = 20$

[PROF: ✓]

(b) Times 10 on every row to kill the decimals:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

[PROF: ✓]

(c) $R_1 \rightarrow R_1 - 2R_3$: $[1 \ 0 \ -4 \mid -40]$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$R_2 \rightarrow R_2 - 3R_1$: $[1 \ 0 \ -4 \mid -40]$

$$[0 \ 7 \ 14 \mid 560]$$

$$[2 \ 1 \ 4 \mid 200]$$

$R_3 \rightarrow R_3 - 2R_1$: $[1 \ 0 \ -4 \mid -40]$

$$[0 \ 7 \ 14 \mid 560]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_2 \rightarrow (1/7)R_2: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 10 \mid 200]$$

$$R_3 \rightarrow (1/10)R_3: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 1 \mid 20]$$

$$x_3 = 20, x_2 = 80 - 2(20) = 40, x_1 = -40 + 4(20) = 40$$

$$\text{Final answer: } x_1 = 40, x_2 = 40, x_3 = 20$$

[PROF: -1 E6 Final answer carries no units; state 40 kg, 40 kg, 20 kg.]

[PROF: Problem 4 7/8]

Problem 5

$$(a) R_2 \rightarrow R_2 - 2R_1: [1 \ 2 \ 1]$$

$$[0 \ 0 \ 1]$$

$$[3 \ 6 \ 4]$$

$$R_3 \rightarrow R_3 - 3R_1: [1 \ 2 \ 1]$$

$$[0 \ 0 \ 1]$$

$$[0 \ 0 \ 1]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 2 \ 1]$$

$$[0 \ 0 \ 1]$$

$$[0 \ 0 \ 0]$$

$$R_1 \rightarrow R_1 - R_2: [1 \ 2 \ 0]$$

$$[0 \ 0 \ 1]$$

$$[0 \ 0 \ 0]$$

$$\text{Final answer: RREF}(A) = [1 \ 2 \ 0; 0 \ 0 \ 1; 0 \ 0 \ 0]$$

[PROF: ✓]

(b) There are pivots in column 1 and column 3 and nothing in row 3, so two pivots.

$$\text{Final answer: rank}(A) = 2$$

[PROF: ✓]

(c) Running the same row operations on $[A \mid b]$ turns the last row into

$$[0 \ 0 \ 0 \mid b_3 - b_1 - b_2].$$

0 solutions is possible: pick $b = [0, 0, 1]^T$, then that last row says $0 = 1$, which is

impossible, so the system is inconsistent.

Exactly 1 solution is impossible: column 2 has no pivot, so x_2 is always free. Any time

the system is consistent that free variable gives a whole family of answers. Put another

way $\text{rank}(A) = 2$ but there are 3 unknowns, and $2 < 3$, so a unique solution can't happen.

Infinitely many is possible: take $b = [1, 3, 4]^T$ (or $b = 0$). Then $b_3 = b_1 + b_2$, the last

row is $0 = 0$, the system is consistent, and free x_2 gives infinitely many solutions.

[PROF: ✓]

[PROF: Problem 5 7/7]

Problem 6

(a) These are dependent because there are three vectors and v_3 has a zero entry, so they

can't fill $\mathbb{R}^3$. The span is a plane.

[PROF: -2 E7 A zero entry in v_3 is not a dependence argument; row reduce to two pivots or give $v_3 = v_2 - 2v_1$.]

(b) None of these is a multiple of another so they are independent; span is $\mathbb{R}^3$.

[PROF: -2 E7 Pairwise non-parallel does not prove independence; show three pivots or $\det = 2 \neq 0$.]

[PROF: Problem 6 4/8]

Grader summary

Problem 1: 3.5/5

Problem 2: 5/6

Problem 3: 5/6

Problem 4: 7/8

Problem 5: 7/7

Problem 6: 4/8

Total: 31.5/40

Problems 3, 4, and 5 are computed correctly and every row operation is labeled.

Problem 6 costs you four points: a zero entry in v_3 and none is a multiple of another are not justifications, and row reduction or an explicit dependence relation is required every time. Show the work in 1(a), round the angle to two decimals, write the general solution parametrically, and give 4(c) in kg.